Project Proposal: Medical Cost Personal (Insurance) Data Analysis

1. Project Definition

This project aims to analyze medical insurance charges using Python. It will explore how age, sex, BMI, number of children, smoking status, and region are associated with insurance costs. The project will include comprehensive data cleaning, exploratory data analysis, at least six statistical visualizations, an interactive User Interface (UI) dashboard with cost prediction, and a short insights report.

2. Dataset Use Case

Dataset: Medical Cost Personal Datasets
Source: Kaggle
Link: https://www.kaggle.com/datasets/mirichoi0218/insurance

The dataset contains 1,338 records and seven columns: age, sex, BMI, children, smoker, region, and charges. It will be used to explore patterns in insurance costs and compare charges across different groups.

3. Planned Visualizations and Expected Outcomes

Graph/Plot	Expected Outcome
Histogram of insurance charges	Understand the distribution and common ranges of charges.
Box plot: smoker vs. non-smoker	Compare typical charges and variability by smoking status.
Scatter plot: age vs. charges	Examine the relationship between age and insurance costs.
Scatter plot: BMI vs. charges	Identify patterns between BMI and insurance costs.
Bar chart: average charges by region	Compare average charges across regions.
Bar chart: average charges by sex	Compare average charges between males and females.

Actual findings will be reported after analyzing the dataset.

4. Python Libraries and Tools

- **Python** — Programming and analysis
- **Pandas** — Data cleaning and manipulation
- **NumPy** — Numerical operations
- **Matplotlib** — Graphs and charts
- **Seaborn** — Statistical visualizations

- **Scikit-Learn** — Linear regression modeling and cost prediction ($R^2 \approx 0.75$)
- **Tkinter & Web UI** — Interactive user interface and dynamic insurance cost calculator
- **Google Colab / Jupyter Notebook** — Code development and execution
- **GitHub** — Project storage and documentation

Data cleaning will include checks for missing values, duplicates, incorrect data types, and potential outliers.

5. GitHub Repository Link

Repository: https://github.com/nandankhagram69/medical-cost-insurance-analysis

The repository will contain the project documentation, proposal, Python notebook, UI application files (Web dashboard & Desktop Tkinter app), and final insights report.

Conclusion

This project will demonstrate the use of Python for data cleaning, exploratory data analysis, visualization, and interactive UI-based cost estimation to identify patterns in medical insurance charges.